Chapter 5

QES potentials and zero energy normalizable states for an extended class of truncated Calogero-Sutherland model

Some ES systems were discussed in the previous chapter. In general, there are few systems for which the Schrödinger equation is exactly sovable and all the eigenvalues and eigenfunctions are completely known. However, several studies have discovered some systems for which a finite number of initial eigenvalues and eigenfunctions are known. These are referred to as QES systems. In quantum mechanics, one of the major problems is obtaining a normalizable zero-energy quantal state corresponding to the Schödinger equation. Quantum states with energy $E = 0$ are generally assumed to belong to the continuum. However, it has been reported in many situations where the zero-energy state can become bound with some restrictions on the potential's parameters [218–222, 227]. In this chapter, we obtain a zero-energy quantal state for three classes of QES potentials corresponding to the rationally extended truncated Calogero-Sutherland model. We use $SO(2,1)$ potential algebra to construct these QES potentials. We find that in all three cases, subjected to some suitable restrictions, the wavefunctions corresponding to $E = 0$ associated with these potentials are normalizable [234].

5.1 The conventional TCS model

The N-body TCS model [261] is defined by the following Hamiltonian

$$\hat{H} = \sum_{i=1}^{N} \left[-\frac{1}{2}\frac{\partial^2}{\partial x_i^2} + \frac{1}{2}\omega^2 x_i^2 \right] + V_{int}, \tag{5.1}$$

here

$$V_{int} = \sum_{\substack{i<j \\ |i-j|\leq r}} \frac{\lambda(\lambda-1)}{\mid x_i - x_j \mid^2} + \sum_{\substack{i<j<k \\ |i-j|\leq r \\ |j-k|\leq r}} \frac{\lambda^2 \mathbf{r}_{ij}.\mathbf{r_{jk}}}{r_{ji}^2 r_{jk}^2}; \qquad \lambda \neq 0, \tag{5.2}$$

in this model, the particles interact via both a pairwise two-body potential and a three-body term. The vector along x-axis is $\mathbf{r}_{ij} = (x_i - x_j)\hat{x}$. The two-body interactions are attractive when $0 < \lambda < 1$ and repulsive when $\lambda \geq 1$. It is important to note that, for the specific cases $r = 1$ and $r = N - 1$, this Hamiltonian reduces to the JK model [262] and the CSM model [263, 264], respectively.
The solution to the above model is provided in [261] and is given by

$$\Psi(\mathbf{x}) = \phi(\mathbf{x})\xi(\mathbf{x}); \qquad \mathbf{x} = (x_1, x_2, ..., x_N)\epsilon\mathbb{R}^N, \tag{5.3}$$

where

$$\phi(\mathbf{x}) = \prod_{i<j}(x_i - x_j)^\lambda \tag{5.4}$$

and the function ξ follows the equation

$$-\frac{1}{2}\sum_{i=1}^{N}\frac{\partial^2\xi}{\partial x_i^2} - \lambda\sum_{i<j}^{N-1}\frac{1}{x_i - x_j}\left(\frac{\partial\xi}{\partial x_i} - \frac{\partial\xi}{\partial x_j}\right) + \left(\frac{1}{2}\sum_{i}^{N}\omega^2 x_i^2 - E\right)\xi = 0. \tag{5.5}$$

To obtain exact solutions to the above equation, one assumes ξ to be

$$\xi = \Phi(\rho)P_s(\mathbf{x}); \qquad \text{where} \quad \rho^2 = \sum_{i=1}^{N} x_i^2. \tag{5.6}$$

By substituting $\xi(\mathbf{x})$ into Eq. (5.5), it can be shown that Φ satisfies the differential equation

$$\Phi''(\rho) + \left(N + 2s - 1 + \lambda r(2N - r - 1)\right)\frac{1}{\rho}\Phi'(\rho) + 2(E - \frac{1}{2}\omega^2\rho^2)\Phi(\rho) = 0\,. \tag{5.7}$$

The function $P_s(\mathbf{x})$ acts as a homogeneous polynomial of degree s (with $s = 0, 1, 2, \dots$) and satisfies a generalized Laplace equation

$$\left[\sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2} + 2\lambda \sum_{i<j}^{N-1} \frac{1}{x_i - x_j}\left(\frac{\partial}{\partial x_i} - \frac{\partial}{\partial x_j}\right)\right] P_s(\mathbf{x}) = 0. \tag{5.8}$$

The solutions to this Laplace equation are explored in detail in Refs. [261, 262, 265]. Additionally, Eq. (5.7) corresponds to the well-known radial equation for the oscillator potential in arbitrary dimensions. The solution to this radial equation is represented by the classical Laguerre orthogonal polynomial $(L_n^{(\alpha)}(\omega\rho^2))$ and is expressed as

$$\Phi(\rho) \simeq \exp(-\frac{\omega\rho^2}{2}) L_n^{(\alpha)}(\omega\rho^2); \quad n = 0, 1, 2, ... \tag{5.9}$$

with the corresponding energy eigenvalues given by

$$E_n = \omega\big(2n + s + \frac{N}{2} + \frac{\lambda r}{2}(2N - r - 1)\big), \tag{5.10}$$

here $\alpha = \big(s - 1 + \frac{N}{2} + \frac{\lambda r}{2}(2N - r - 1)\big)$. As shown in [261], when $r = 1$ and $r = N - 1$, the results correspond to those of the JK model [262] and the CSM model [263, 264], respectively.

5.2 Extended truncated Calogero-Sutherland model

The extended N-body TCS model in one dimension involves N particles that are harmonically confined and interact pairwise via a two-body potential, with an additional three-body interaction term. The concept of truncating the interaction focuses on the number of neighboring particles instead of their relative distances. The Hamiltonian of interest is given by the form

$$H_{ext} = \hat{H} + V_{new} \tag{5.11}$$

where, using the notations of [148, 261]

$$\hat{H} = \sum_{i=1}^{N} \left[-\frac{1}{2}\frac{\partial^2}{\partial x_i^2} + \frac{1}{2}\omega^2 x_i^2 \right] + V_{int}, \tag{5.12}$$

$$V_{int} = \sum_{\substack{i<j \\ |i-j|\leq r}} \frac{\lambda(\lambda-1)}{|\, x_i - x_j \,|^2} + \sum_{\substack{i<j<k \\ |i-j|\leq r \\ |j-k|\leq r}} \frac{\lambda^2 \mathbf{r}_{ij}.\mathbf{r_{jk}}}{r_{ji}^2 r_{jk}^2}; \qquad \lambda \neq 0, \tag{5.13}$$

and V_{new} is an additional interaction term defined by

$$V_{new} = \frac{(\alpha_1 + \alpha_2\omega^2\rho^2)}{(\beta_1 + \beta_2\omega^2\rho^2)^2}, \tag{5.14}$$

which contains some unknown coefficients. The variable ρ in Eq. (5.14) varies from 0 to ∞, unlike the variable x in the $SO(2,1)$ potential algebra, which varies from $-\infty$ to ∞.

To find the solution of the Schrödinger equation associated with Eq. (5.11),

$$\hat{H}_{ext}\Psi_{ext} = E_{ext}\Psi_{ext}. \tag{5.15}$$

We consider $\Psi_{ext}(\mathbf{x}) = \phi(\mathbf{x})\xi_{ext}$, where $\phi(\mathbf{x})$ is given by $\Pi_{i<j}(x_i - x_j)^\lambda$, and ξ_{ext} satisfying

$$\begin{aligned} &-\frac{1}{2}\sum_{i=1}^{N}\frac{\partial^2\xi_{ext}}{\partial x_i^2} - \lambda\sum_{i<j}^{N-1}\frac{1}{x_i - x_j}\left(\frac{\partial\xi_{ext}}{\partial x_i} - \frac{\partial\xi_{ext}}{\partial x_j}\right) \\ &+\left(\frac{1}{2}\sum_{i}^{N}\omega^2 x_i^2 + V_{new} - E_{ext}\right)\xi_{ext} = 0. \end{aligned} \tag{5.16}$$

Like conventional model, continuing with the redefinition $\xi_{\text{ext}} = \Phi_{\text{ext}}(\rho)P_s(\mathbf{x})$, we observe that $\Phi_{\text{ext}}(\rho)$ satisfies the radial equation [148]

$$\begin{aligned} &\Phi''_{ext}(\rho) + \big(N + 2s - 1 + \lambda r(2N - r - 1)\big)\frac{1}{\rho}\Phi'_{ext}(\rho) \\ &+ 2\big(E - (\frac{1}{2}\omega^2\rho^2 + V_{new})\big)\Phi_{ext}(\rho) = 0, \end{aligned} \tag{5.17}$$

where $P_s(\mathbf{x})$ satisfies the Laplace equation. In above Eq. (5.17), a prime on $\Phi_{\text{ext}}(\rho)$ signifies differentiation with respect to ρ. To solve it, we proceed with the standard approach [266] of substituting

$$\Phi_{ext}(\rho) = f(\rho)\zeta(g(\rho)), \tag{5.18}$$

where $f(\rho)$ denotes an unknown coefficient function that needs to be determined, $g(\rho)$ represents a variable transformation, and $\zeta(g)$ is selected based on the requirement that it satisfies a second-order differential equation

$$\zeta''(g(\rho)) + Q_1(g)\zeta'(g(\rho)) + R_1(g)\zeta(g(\rho)) = 0. \tag{5.19}$$

such that the functions $Q_1(g)$ and $R_1(g)$ are well-defined if a specific function is choosen for it. Substituting Eq. (5.18) into Eq. (5.17), we obtain

$$\zeta''(g) + \left(\frac{2f'(\rho)}{f(\rho)g'(\rho)} + \frac{g''(\rho)}{g'(\rho)^2} + \frac{\tau}{\rho g'(\rho)}\right)\zeta'(g) + \frac{1}{g'(\rho)^2}\left(\frac{f''(\rho)}{f(\rho)} + \frac{\tau f'(\rho)}{\rho f(\rho)} + 2(E_{ext} - V_{ext})\right)\zeta(g) = 0, \tag{5.20}$$

here $V_{ext} = \frac{1}{2}\omega\rho^2 + V_{new}$ and $\tau = \big(N + 2s - 1 + \lambda r(2N - r - 1)\big)$. On comparing Eq. (5.20) with Eq. (5.19), we find

$$\begin{aligned} Q_1(g) &= \frac{2f'(\rho)}{f(\rho)g'(\rho)} + \frac{g''(\rho)}{g'(\rho)^2} + \frac{\tau}{\rho g'(\rho)} \text{ and} \\ R_1(g) &= \frac{1}{g'(\rho)^2}\left(\frac{f''(\rho)}{f(\rho)} + \frac{\tau f'(\rho)}{\rho f(\rho)} + 2(E_{ext} - V_{ext})\right). \end{aligned} \tag{5.21}$$

Upon simplifying $Q_1(g)$, it becomes evident that

$$f(\rho) \simeq (g'(\rho))^{-\frac{1}{2}}\rho^{-\frac{\alpha}{2}} \exp\left(\frac{1}{2}\int^g Q_1(g)dg\right). \tag{5.22}$$

Using it in the expression for $R_1(g)$, we obtain the expression

$$\begin{aligned} E_{ext} - V_{ext} =& \frac{1}{2}\left[\frac{g'''(\rho)}{2g'(\rho)} - \frac{3}{4}\frac{g''(\rho)^2}{g(\rho)^2} + \frac{\tau/2(\tau/2-1)}{\rho^2}\right. \\ &\left. + g'(\rho)^2\left(R_1(g) - \frac{Q_1'(g)}{2} - \frac{Q_1^2(g)}{4}\right)\right]. \end{aligned} \tag{5.23}$$

This can be further simplified as

$$\begin{aligned} E_{ext} - V_{ext} =& \frac{1}{2}\left[\frac{\Delta V(\rho)}{2} + \frac{\tau/2(\tau/2-1)}{\rho^2}\right. \\ &\left. + g'(\rho)^2\left(R_1(g) - \frac{Q_1'(g)}{2} - \frac{Q_1^2(g)}{4}\right)\right]. \end{aligned} \tag{5.24}$$

where the $\Delta V(\rho)$ represents the Schwartzian derivative,

$$\Delta V(\rho) = \frac{g'''(\rho)}{g'(\rho)} - \frac{3}{2}\frac{g''(\rho)^2}{g(\rho)^2}. \tag{5.25}$$

Equation (5.24) is the key relation that we will utilize below to search for normalizable zero-energy solutions within an extended class of the truncated Calogero-Sutherland model.

5.3 Normalizable zero-energy quantum states

To identify QES systems with normalizable zero-energy solutions, we follow the procedure outlined in [227]. To begin, we set $Q_1(g) = 0$, leading to the condition

$$f(\rho)^2 g'(\rho)\rho^\tau = \text{constant}. \tag{5.26}$$

This simplifies equation (5.24) to the form

$$E_{ext} - V_{ext} = \frac{\Delta V(\rho)}{4} + \frac{\tau/4(\tau/2-1)}{\rho^2} + \frac{1}{2}g'(\rho)^2 R_1(g). \tag{5.27}$$

Next, to obtain a meaningful representation of (5.27), we set $R_1 = E_T - V_T(g)$ and employ the transformation $\rho = K(g(\rho))$, yielding

$$E_T - V_T(g) = [E_{ext} - V_{ext}]2K'(\rho))^2 + \frac{1}{2}\Delta V(K(g)) - \frac{\frac{\tau}{2}(\frac{\tau}{2}-1)}{K(g)^2}K'(g)^2. \tag{5.28}$$

Now, utilizing $SO(2,1)$ as the potential algebra of the Schrödinger equation, and setting $V_T = V_m$, gives

$$\begin{aligned}&\left(\frac{1}{4} - m^2\right)(1-F^2) - 2mFG + G^2 - E_T \\ &= 2[K'(g)]^2[V_{ext}(K(g)) - E_{ext}] - \frac{1}{2}\Delta V(K(g)) + \frac{\frac{\tau}{2}(\frac{\tau}{2}-1)}{K(g)^2}K'(g)^2.\end{aligned} \tag{5.29}$$

Now, considering the first class of solutions (2.117), using the mapping function $\rho = K(g) = \frac{1}{e^g-1}$ and setting $E = 0$, yields

$$\begin{aligned}V(K(g)) = &\left(1 - 4m^2 + 4b^2\right) \text{sech } g(\text{sech } g - 2) - 8mb(\tanh g \text{ sech } g \\ &- 2\tanh g + \sinh g) - \left(4E_T + \frac{1}{2}\right)\cosh g(\cosh g - 2) \\ &- 4\left(E_T + m^2 - b^2 - \frac{1}{8}\right) + \frac{\tau}{4}(\frac{\tau}{2} - 1)(2\sinh^2 g + \sinh g).\end{aligned} \tag{5.30}$$

Thus, by utilizing the mapping function and the above equation, we can determine $V(\rho)$

$$\text{(I)} \qquad \begin{aligned}V(\rho) = &-\frac{A}{(2\rho^2+2\rho+1)^2} - B\frac{2\rho+1}{\rho(\rho+1)\,(2\rho^2+2\rho+1)^2} \\ &+ \frac{C}{\rho^2(\rho+1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2}-1)}{\rho^2},\end{aligned} \tag{5.31}$$

where

$$\begin{aligned} A =& \frac{1}{2}(4\left(m^2-b^2\right)-1), \qquad B = 2mb, \\ C =& -\frac{1}{2}(E_T+\frac{1}{4}) = \left(k-\frac{1}{2}\right)^2 - \frac{1}{4} = k(k-1). \end{aligned} \tag{5.32}$$

The initial example of a QES potential with a known $E = 0$ is provided in equation (5.31). For non-negative values of C and with $\tau = 0$, its behavior at the origin exhibits similarities to the (shifted) Coulomb effective potential $V(\rho) = \frac{\hbar^2}{2m}\frac{l(l+1)}{\rho^2} - \frac{e^2}{\rho}$. The wave function corresponding to the first-class potential with $m = k$ is given by $\psi(\rho) = \sqrt{K'(g(\rho))}\chi_0$, where $\chi_0 \sim G^{k-\frac{1}{2}}h$ represents the ground state wave function of the class I potential in Eq. (2.117), with $h = \exp\left[b\tan^{-1}(\sinh g)\right]$ [247]. This yields

$$\psi(\rho) = 2^{k-\frac{1}{2}}\sqrt{\rho(\rho+1)}\rho^{-\frac{\tau}{2}}\left(\frac{b\rho(\rho+1)}{2\rho(\rho+1)+1}\right)^{k-\frac{1}{2}} e^{b\tan^{-1}\left(\frac{2\rho+1}{2\rho^2+2\rho}\right)}. \tag{5.33}$$

From the above expression, it is evident that as $\rho \to \infty$, the expression depends solely on $\rho^{1-\frac{\tau}{2}}$. For the wavefunction to be well-behaved, the exponent must be negative. This condition requires that $\tau > 2$. Conversely, as $\rho \to 0$, the expression depends on $\rho^{k-\frac{\tau}{2}}$. In this case, for the wavefunction to be well-behaved, the exponent $k - \frac{\tau}{2}$ must be positive. This imposes the constraint that $2k > \tau$. In summary, the wavefunction is well-behaved as long as the constraints $2k > \tau$ and $\tau > 2$ are satisfied. Figure 5.1 presents the plots of the probability density and wave function for various values of k, τ, and b.

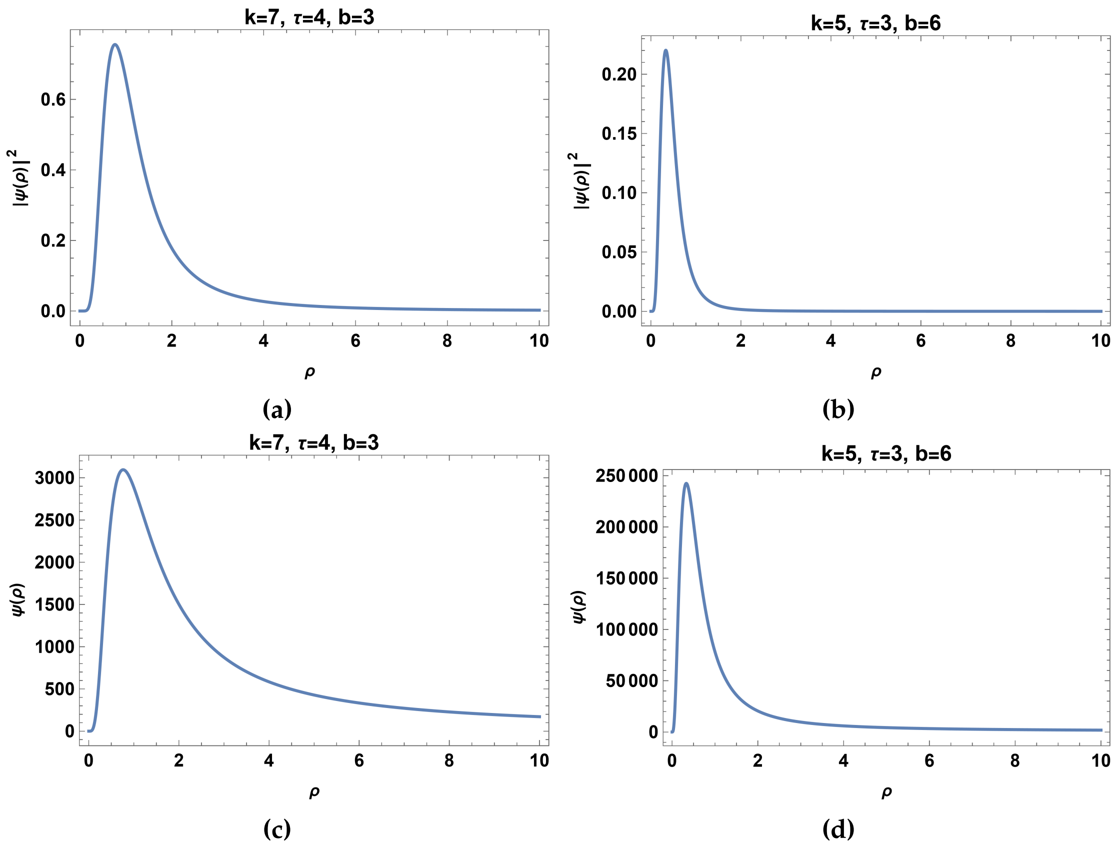

Figure 5.1: (a) & (b) The plot illustrates the probability density for the first class potential at different values of k, τ, and b (c) & (d) The plot illustrates the wavefunction for the first class potential at different values of k, τ, and b.

Similarly, we can identify the QES potentials for the other two classes

$$\text{(II)} \qquad V(\rho) = \frac{b^2}{2(\rho+1)^4} - \frac{mb}{\rho(\rho+1)^3} - \frac{E_T + \frac{1}{4}}{2\rho^2(\rho+1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2}-1)}{\rho^2}. \tag{5.34}$$

The zero-energy wave function for the second-class potential with $m = k$ is given by $\psi(\rho) = \sqrt{K'(g(\rho))}\chi_0$, where $\chi_0 \sim G^{k-\frac{1}{2}}h$ and $h = \exp\left[-be^{-g}\right]$. This leads to

$$\psi(\rho) = \sqrt{\rho(\rho+1)}e^{-\frac{b\rho}{\rho+1}}\rho^{-\frac{\tau}{2}}\left(\frac{b\rho}{\rho+1}\right)^{k-\frac{1}{2}}. \tag{5.35}$$

The wave function $\psi(\rho)$ exhibits acceptable asymptotic behavior under the conditions $2k > \tau$, $\tau \geq 4$, and $b > 0$. A graphical representation of the probability density and the wavefunction is depicted in Figure (5.2) for various values of k, τ, and b.

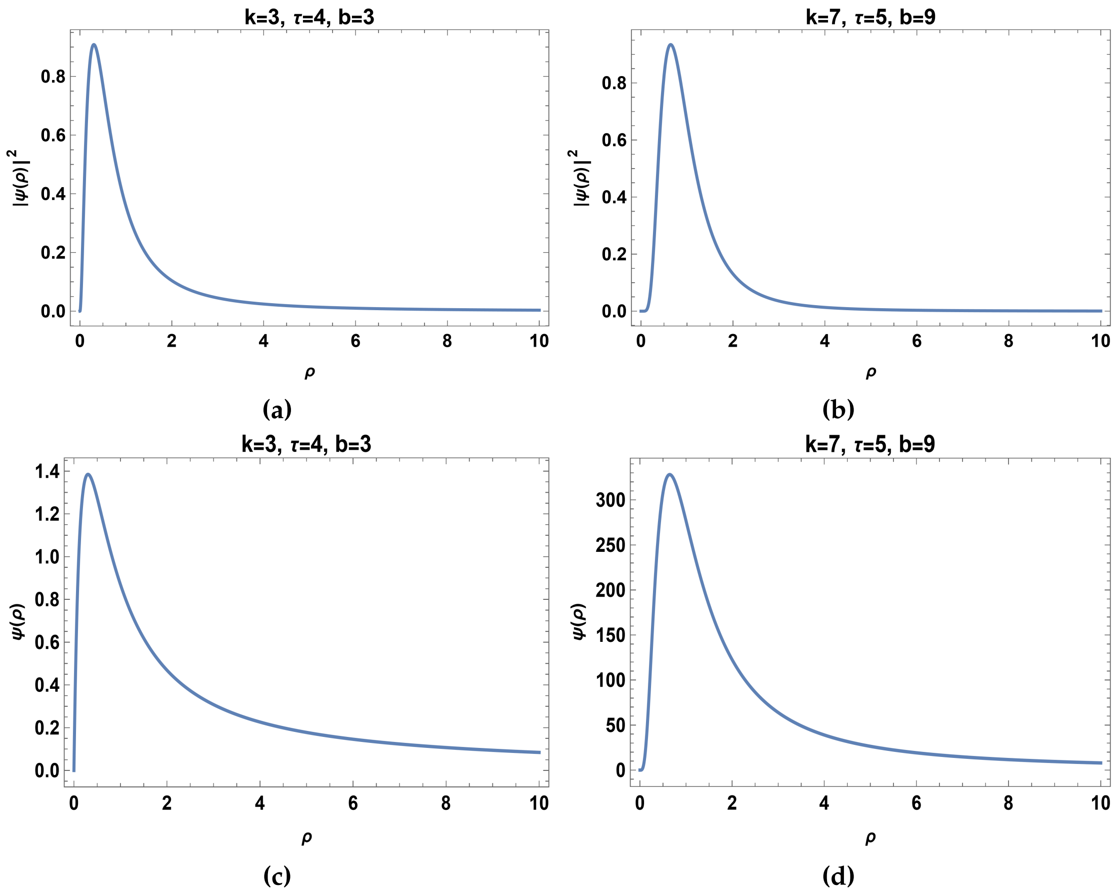

Figure 5.2: (a) & (b) The plot illustrates the probability density for the second class potential at different values of k, τ, and b (c) & (d) The plot illustrates the wavefunction for the second class potential at different values of k, τ, and b.

The third type of QES potential takes the form

$$\text{(III)} \qquad V(\rho) = \frac{4(m+b)^2 - 1}{2(2\rho+1)^2} - \frac{2mb}{\rho(\rho+1)} - \frac{E_T + \frac{1}{4}}{2\rho^2(\rho+1)^2} - \frac{\frac{\tau}{4}(\frac{\tau}{2}-1)}{\rho^2}. \tag{5.36}$$

The corresponding wave function for this potential with $m = k$ is given by

$$\psi(\rho) = 2^{k-\frac{1}{2}}\sqrt{\rho(\rho+1)}\left(\frac{1}{2\rho+1}\right)^{b}\rho^{-\frac{\tau}{2}}\left(\frac{b\rho(\rho+1)}{2\rho+1}\right)^{k-\frac{1}{2}}. \tag{5.37}$$

The normalization integral is expressed as

$$\int_0^\infty |\psi|^2 d\rho = \int_0^\infty \frac{\rho^{-\tau}[\rho(\rho+1)]^{2k}}{(2\rho+1)^{2k+2b-1}} d\rho. \tag{5.38}$$

We investigate its convergence using the criterion stating that $\int_a^\infty f(\rho)d\rho$ diverges if $\lim_{\rho\to\infty}\rho^\alpha f(\rho) = L \neq 0$ and $\alpha \leq 1$. In this scenario, as $\rho \to \infty$, $f(\rho) \sim \rho^{2k-2b-\tau+1}$. By setting $\alpha = 2b - 2k + \tau - 1$, we determine that $\lim_{\rho\to\infty}\rho^\alpha f(\rho) = L \neq 0$. The integral converges if $2b - 2k + \tau - 1 > 1$, which simplifies to $b > k - \frac{\tau}{2} + 1$. Conversely, it diverges if $b \leq k - \frac{\tau}{2} + 1$.

It appears to be well-behaved under the conditions $2k > \tau$ and $b > k - \frac{\tau}{2} + 1$. The plots of the probability density and the wavefunction are depicted in Figure (5.3) for various values of k, τ, and b. Graphs illustrating the potential for various parameter choices are presented in Figure (5.4).

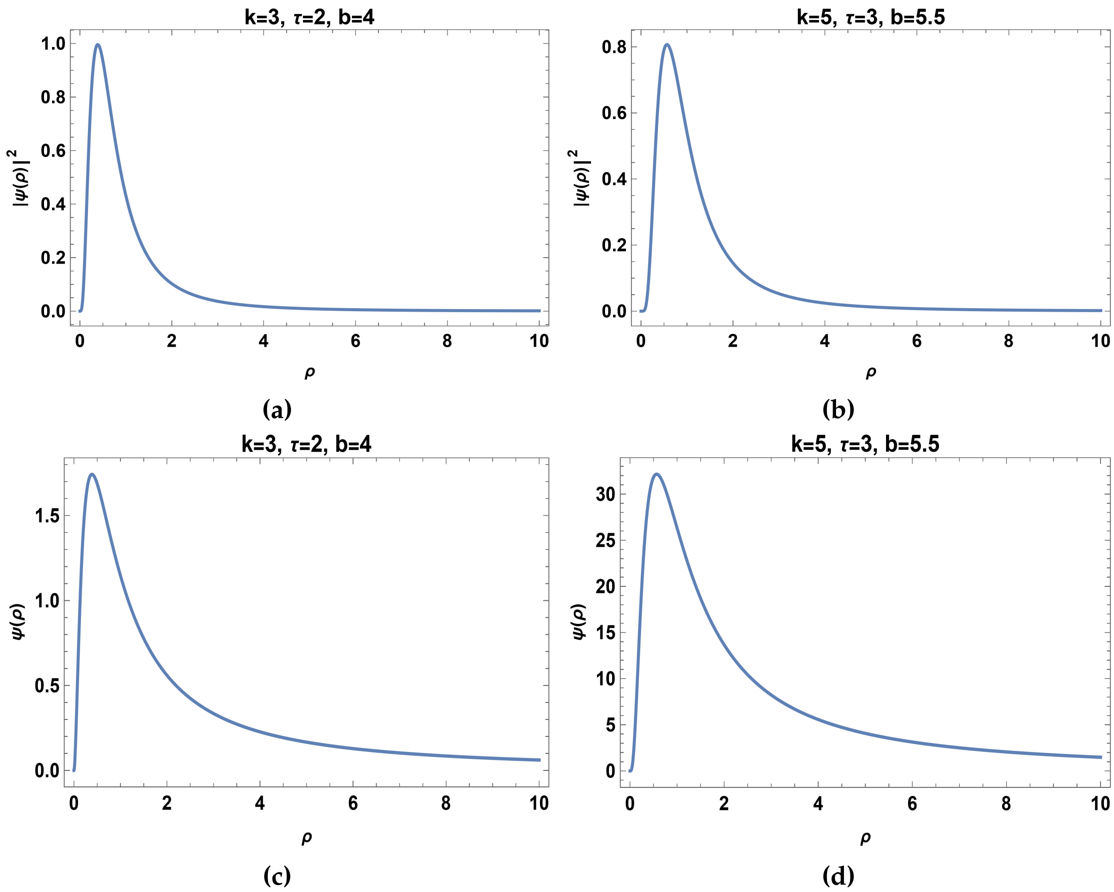

Figure 5.3: (a) & (b) The plot illustrates the probability density for the third class potential at different values of k, τ, and b (c) & (d) The plot illustrates the wavefunction for the third class potential at different values of k, τ, and b.

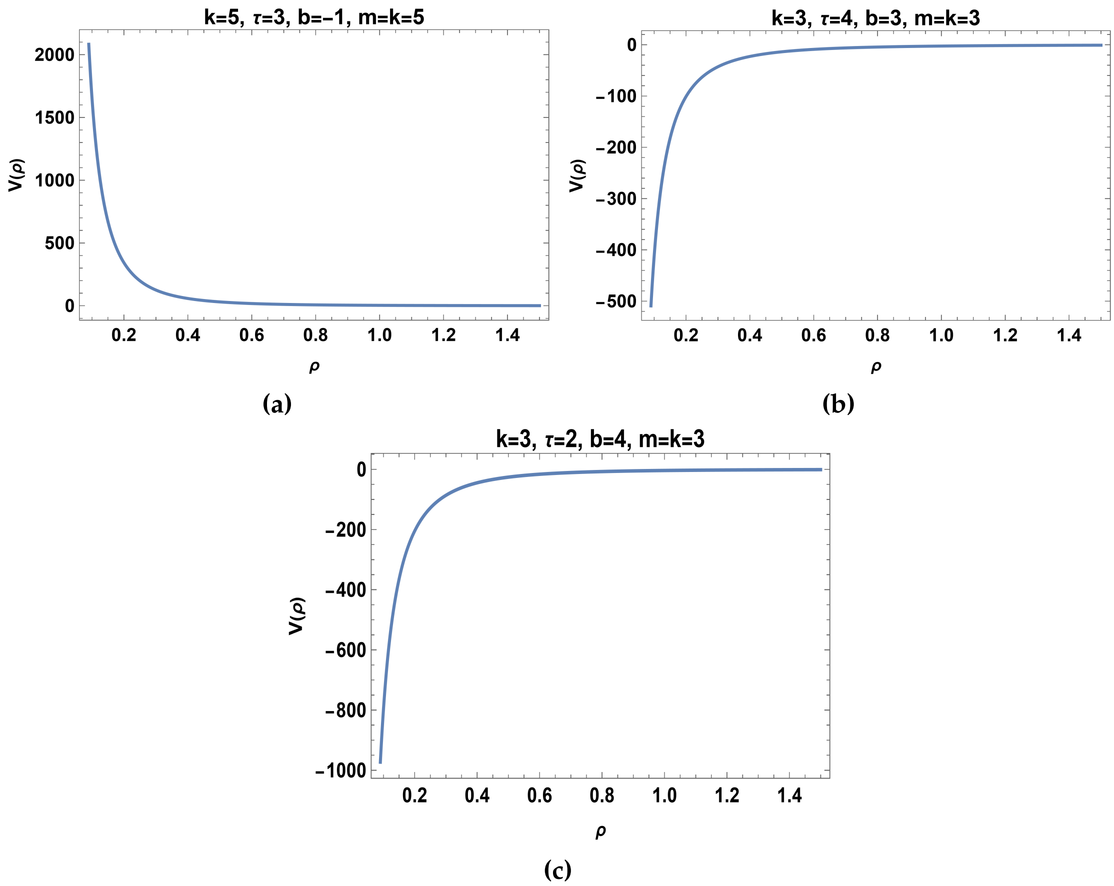

Figure 5.4: (a) The plots of potential $V(\rho)$ of equation (5.31). (b) The plot of potential $V(\rho)$ of equation (5.34). (c) The plot of potential $V(\rho)$ of equation (5.36).

5.4 Result and discussion

We have considered three cases corresponding to the constraints on the guiding functions of $SO(2,1)$. An important result of our analysis is that in all three cases, we identified normalizable wavefunctions by imposing reasonable constraints on the coupling parameters. This contrasts sharply with a similar situation encountered in a previous study [227] when addressing a class of QES rational potentials with known $E = 0$ eigenvalue. It was found that the wave function of the zero-energy state for only one class of potentials turned out to be normalizable under certain convergence condition being satisfied. In the study of a rationally extended truncated Calogero-Sutherland model, we applied a point canonical transformation to the associated Schrödinger equation and equated the resulting energy expression with the Casimir operator of the $SO(2,1)$ algebra. This approach led to the identification of three new classes of QES rational potentials corresponding to the $E = 0$ state. For each of these potentials, we demonstrated that the solutions to the eigenvalue equation at zero energy were both normalizable and asymptotically smooth, under appropriate restrictions on the coupling parameters.